Design and Implementation of a General-Purpose Classical AI Logical Reasoning System

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Abstract

This report presents an end-to-end symbolic reasoning system that decides whether a query statement is logically entailed by a knowledge base of facts and rules. Parsing is separated from inference, and every decision is made by a classical reasoning engine; no generative model produces a prediction. The system supports three representations: first-order Horn clauses, deterministic structural causal models (SCMs) and full first-order logic (FOL) syntax trees. It implements forward chaining, backward chaining with unification, A* proof search, Pearl's abduction–action–prediction procedure, and Herbrand-grounded DPLL refutation. On the 3,829 public queries from RuleTaker, CounterBench and FOLIO it obtains 98.5% on RuleTaker, 96.0% on CounterBench and 78.0% on FOLIO (entailed vs. not entailed), with a total execution time of about 1.3 s.

1 Chosen Internal Representation

Input assumption. Following the assignment instructions, the translation from natural language into logic is not the focus of this work. Rule-chaining and causal inputs are parsed with deterministic templates, and for FOLIO the system consumes the supplied first-order annotations (`premises-FOL`, `conclusion-FOL`) as its input logic.

1.1 First-order Horn clauses and atomic facts

For RuleTaker and syllogistic knowledge bases, the knowledge is a set of typed terms (constants c and variables $?x$), atoms $P(t_1, \ldots, t_k)$ and Horn rules

$$P_1(x) \wedge P_2(x) \wedge \cdots \wedge P_m(x) \rightarrow Q(x). \tag{1}$$

Templates such as "Based on A, it is evident that B", "When A, we can conclude B" and "If someone is P and Q, then they are R" are parsed by regular expressions into ground facts and quantified Horn clauses. Queries are answered under the closed-world assumption: an atom that cannot be derived is `False`.

1.2 Structural causal equations (CounterBench)

Deterministic counterfactual questions are represented as assignments in a directed causal graph $G = (V, E)$:

$$v_i := f_i(\mathrm{Pa}(v_i)), \qquad f_i \in \{\wedge, \vee, \neg\}. \tag{2}$$

Statements such as "A and B together cause not C", "V or not T causes S and F" and "A causes B, C and D" are decomposed into one equation per target; a multi-target list inherits the antecedent of the preceding clause. Observations ("We observed X") are stored in an observation map O. Parentless variables that are not observed default to `False`.

1.3 First-order logic syntax trees (FOLIO)

Formulas using $\forall$, $\exists$, $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$ and $\oplus$ are parsed into a recursive abstract syntax tree. A string-similarity pass (`SequenceMatcher`) aligns predicate and constant names in the conclusion with those of the premises, to absorb small spelling differences between the two annotations.

2 Classical Reasoning Algorithms

All entailment decisions are made by symbolic inference. Forward chaining (RuleTaker), the causal engine (CounterBench) and DPLL refutation (FOLIO) form the evaluated path; backward chaining is used for the natural-language fallback on general knowledge bases, and A* proof search is implemented as an alternative prover but is not used in the benchmark runs.

1. **Forward chaining with Robinson unification.** Starting from the facts F_0, rule antecedents are unified with known ground facts until the query is derived or a fixed point $F_{t+1} = F_t$ is reached.
2. **Backward chaining (DFS with cycle detection).** For a goal q, the engine searches rules whose consequent unifies with q and recursively proves the antecedents, keeping a visited-goal set so cyclic rule graphs terminate.
3. **A* proof search.** A state s is the set of unproven subgoals, $g(s)$ is the number of inference steps and $h(s) = |s \setminus F_0|$ estimates the remaining work.
4. **Abduction–action–prediction.** (i) *Abduction* fixes the observed exogenous variables to their observed values (observations of derived variables are not used); (ii) *action* applies the intervention $\mathrm{do}(X = x)$ by overriding the equations of the intervened variables (several interventions are applied together for joint and nested questions); (iii) *prediction* propagates truth values to evaluate the query.
5. **Herbrand grounding and DPLL refutation.** Quantified formulas are instantiated over a bounded Herbrand universe. KB $\wedge \neg$Query is converted to CNF by a memoized Tseitin transformation and solved by a step-bounded DPLL solver with unit propagation. If it is unsatisfiable (and the KB is satisfiable), the query is entailed. The domain is capped at three constants (two inside nested quantifiers; those of the conclusion first) and the search stops after 100 nodes, treating the formula as satisfiable, i.e. the query as not entailed.

3 Experiments on the Public Datasets

Label convention. RuleTaker and CounterBench are binary. FOLIO has three labels (`True`, `False`, `Uncertain`); because the task asks for entailment, the system outputs `True` only if the conclusion is entailed and `False` otherwise, so gold labels `False` and `Uncertain` are both treated as "not entailed". Under strict three-way matching the FOLIO accuracy would be lower (46.2% on train, 50.2% on validation), since the system never outputs `Uncertain`.

3.1 Discussion

- **RuleTaker (97.6–99.5%).** Forward chaining answers all 1,625 queries in about 0.15 s. All 25 errors are false positives: the query is derivable from the rules, but the dataset label is `False`. A likely cause is label noise from the dataset's generator, which may not have registered every derivable rule, although we did not verify this.
- **CounterBench (96.0%).** The structural engine answers 1,000 questions in 0.06 s (about 0.06 ms per question). Accuracy by type is: basic 250/250, conditional 230/250, joint

Table 1: Accuracy and execution time on the public benchmarks (times from one run on the evaluation machine; they vary slightly between runs).

Split	Queries	Correct	Accuracy	Time (s)
`ruletaker_train`	1,000	983	98.30%	0.0916
`ruletaker_test`	250	244	97.60%	0.0229
`ruletaker_ood_test`	375	373	99.47%	0.0376
`counterbench_alpha`	1,000	960	96.00%	0.0609
`FOLIO_train`	1,001	774	77.32%	0.8794
`FOLIO_validation`	203	165	81.28%	0.2014
Overall	3,829	3,499	91.38%	1.2941

240/250, nested 240/250. The remaining errors occur in graphs with observed or intervened variables on negated or "or" branches; we did not isolate a single cause.

- **FOLIO (77.3% train, 81.3% validation).** Always answering `False` would score 61.2% (train) and 64.5% (validation), so the solver is clearly better than the majority baseline. Its errors are asymmetric (Table **??**): it rarely asserts an entailment that does not hold, but misses many true entailments, consistent with the three-constant domain cap and the 100-node DPLL budget, both of which push undecided cases towards `False`.

Table 2: FOLIO breakdown (entailed vs. not entailed).

Split	Recall (gold `True`)	Recall (not `True`)	Majority baseline
Train	190/388 (49.0%)	584/613 (95.3%)	61.2%
Validation	41/72 (56.9%)	124/131 (94.7%)	64.5%

4 Examples of Reasoning Traces

Trace 1: unification and forward chaining

KB: Bob is young. Bob is kind. If someone is young, then they are smart. If someone is smart and kind, then they are successful. **Query:** Bob is successful.

- $F_0 = \{\text{young(bob)}, \text{kind(bob)}\}$.
- Step 1: unify young($?x$) with young(bob), $\theta = \{?x \mapsto \text{bob}\}$; derive smart(bob).
- Step 2: unify $\text{smart}(?x) \wedge \text{kind}(?x)$ with $\{\text{smart(bob)}, \text{kind(bob)}\}$; derive successful(bob).
- Prediction: `True`.

Trace 2: RuleTaker OOD, depth 4

Facts: Henry reads books. Sarah goes to the hospital. Henry carries bags. **Rules:** R1: henry carries bags → the teacher carries bags; R2: henry carries bags → the teacher reads books; R3: the teacher reads books → sarah carries bags. **Query:** Sarah carries bags.

- Step 1: *henry carries bags* $\in F_0$ triggers R2 and adds *the teacher reads books*.
- Step 2: that fact triggers R3 and adds *sarah carries bags*.
- Step 3: the goal is in F_2. Prediction: `True`.

Trace 3: CounterBench counterfactual (question 1)

Given: Ziklo causes Blaf, Blaf and Ziklo together cause Trune, Trune causes not Vork, Vork causes Lumbo. **Query:** Would Lumbo occur if not Ziklo instead of Ziklo?

- Equations: $\mathit{blaf} := \mathit{ziklo}$; $\mathit{trune} := \mathit{blaf} \wedge \mathit{ziklo}$; $\mathit{vork} := \neg \mathit{trune}$; $\mathit{lumbo} := \mathit{vork}$.
- Intervention: $\mathrm{do}(\mathit{ziklo} = \mathrm{False})$.
- Propagation: $\mathit{blaf} = \mathrm{False} \Rightarrow \mathit{trune} = \mathrm{False} \Rightarrow \mathit{vork} = \mathrm{True} \Rightarrow \mathit{lumbo} = \mathrm{True}$.
- Prediction: `True` (yes).

Trace 4: FOLIO refutation (example 1175)

Premises: $\forall x\,(\mathrm{Have}(x, \mathrm{typecport}) \rightarrow \neg\mathrm{ProducedBefore}(x, \mathrm{yr2010}))$; $\forall x\,(\mathrm{Monitor}(x) \wedge \mathrm{In}(x, \mathrm{library}) \rightarrow \mathrm{ProducedBefore}(x, \mathrm{yr2010}))$; $\mathrm{Monitor}(\text{l-2021}) \wedge (\mathrm{In}(\text{l-2021}, \mathrm{library}) \oplus \mathrm{Have}(\text{l-2021}, \mathrm{typecport}))$; $\neg(\mathrm{ProducedBefore}(\text{l-2021}, \mathrm{yr2010}) \oplus \mathrm{ProducedBy}(\text{l-2021}, \mathrm{lg}))$. **Conclusion:** $\mathrm{Have}(\text{l-2021}, \mathrm{typecport}) \oplus \mathrm{ProducedBy}(\text{l-2021}, \mathrm{lg})$.

- Asserting the negated conclusion forces Have $\leftrightarrow$ ProducedBy. Together with ProducedBefore $\leftrightarrow$ ProducedBy, unit propagation gives Have $\leftrightarrow$ ProducedBefore.
- If Have is true this contradicts the first premise; if it is false it contradicts premises 2 and 3. The empty clause $\square$ is derived.
- Prediction: `True`.

5 Limitations

Inference is closed-world, so non-derivable statements are labelled `False`; FOLIO `Uncertain` cannot be distinguished from `False`. The DPLL solver is step-bounded, which trades recall of true entailments for speed.